

# ARNAV JOSHI

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## Education

**Thapar Institute of Engineering and Technology**

Aug 2023 – July 2027

*B.Tech in Computer Science Engineering*

**CGPA: 9.57**

**Ryan International School, Ludhiana**

July 2023

*Class 12 (Senior Secondary Education)*

**96.7%**

## Experience

**Samsung R&D Institute, Bengaluru**

July 2026 – Dec 2026

*Research Intern | Cloud AI/ML*

- Building a Retrieval-Augmented Generation (RAG) pipeline for an AI-powered cloud-operations and in-built assistant, enriching natural-language queries with application-specific context.
- Designing an agentic decision layer that lets users accept/reject AI-recommended actions, orchestrating multiple LLM backends into one troubleshooting interface.

**Indian Institute of Technology Delhi — FE<sup>2</sup>B Lab**

May 2026 – July 2026

*AI/ML Intern | Advisor: Dr. Narsing Kumar Jha*

- Designed a feature-engineering and unsupervised clustering pipeline to classify bubble-acoustic flow regimes from 90 real hydrophone recordings, achieving 95.56% cross-validated accuracy.
- Built a physics-informed neural network (PINN-inspired) regression model fusing PIV velocity fields with force-balance priors to predict dipole acoustic pressure and full 360° directivity, validated against 110 real experimental frames.

## Projects

**SynthSales – Agentic AI Sales CRM** | *Python, FastAPI, PostgreSQL, LLM APIs*

GitHub | Live

- Architected an 8-agent LLM pipeline for automated company research, scoring, and outreach, with a confidence-calibrated ranking layer that hard-caps AI-driven scores by up to 54 points when evidence is weak.
- Engineered a 3-provider LLM failover chain (Gemini/Groq/OpenRouter) with automatic rate-limit rerouting, self-correcting structured generation, and a 6-class intent classifier gating replies at a 70% confidence threshold.

**Physics-Informed Halo CME Detection System** | *Python, XGBoost, Scikit-learn, CDF Ingestion*

GitHub

- Built a physics-informed ML pipeline detecting Halo CMEs from Aditya-L1 SWIS solar-wind data, deriving 4 domain-specific features from 8 candidates via class-separability screening.
- Trained a soft-voting ensemble (Random Forest, XGBoost, Logistic Regression) on 43 labeled event windows, reaching 85% accuracy and zero false negatives; deployed as a public Streamlit app.

**Sentinel-RAG – Guarded Enterprise RAG Assistant** | *LangGraph, NeMo Guardrails, Qdrant, RAGAS*

GitHub

- Built a two-stage prompt-injection cascade (Llama Prompt Guard 2 + LLM adjudicator) that separates benign engineering phrasing from persona-hijack attacks, achieving 1.000 precision/recall on held-out adversarial data.
- Ran an independent-judge RAGAS evaluation suite across guardrails, retrieval, and generation, delivering 100% hit@5 retrieval accuracy and 0.917 answer relevancy against a 58-document distractor corpus.

## Skills

**Programming Languages:** Python, C++, JavaScript, SQL, Go

**Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, PINNs, Feature Engineering, Signal Processing

**NLP & Generative AI:** HuggingFace Transformers, Wav2Vec2, LLM APIs, LangChain, LangGraph, RAG, Vector Databases (Qdrant), Agentic AI Systems, Stable Diffusion, NLTK

**MLOps & Deployment:** MLflow, DVC, Docker, Kubernetes, AWS EC2, GitHub Actions, FastAPI, Flask, Streamlit

**Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, OpenCV, MongoDB, Firebase

## Achievements

- Achieved a Top 1500 global ranking in the Google Big Code Competition and was selected as a National Semi-Finalist in Flipkart GRiD 8.0.
- Awarded Merit Scholarship for three consecutive years at Thapar Institute of Engineering and Technology, ranking among the top students of the Computer Science branch.